THE COMPARISON MAP BETWEEN PULLBACK AND THE RIGHT ADJOINT IS AN ISOMORPHISM FOR A PERFECT FIRST FACTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9T (sources/stacks-duality/pinned-original-source.tex line 2085).

Allowed standard prerequisites (editorial).

- The dual of a perfect complex. Cohomology, Lemma labelled cohomology-lemma-dual-perfect-complex, cited at proof lines 2118-2120. Provides K^vee and the left adjunction step.
- The projection formula and the base change version. Cohomology, Lemma labelled cohomology-lemma-projection-formula-perfect, cited at proof lines 2124-2128, and Derived Categories of Schemes, Lemma labelled perfect-lemma-cohomology-base-change, cited at proof lines 2126-2130. Gives the middle equality.
- Adjunction defining the right adjoint a. Same chapter, Lemma labelled lemma-twisted-inverse-image, invoked in the proof at proof lines 2120-2124. Gives the adjunction step.

Source reference. Original equation 0A9S: sources/stacks-duality/pinned-original-source.tex, source lines 2079–2079. Same-chapter pointer into the author environment equation at source line 2079; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2097 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0B54: sources/stacks-chow/tags.txt, source lines 14022–14022. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2112 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08EU: sources/stacks-chow/tags.txt, source lines 10685–10685. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2114 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. The map $Lf^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} a(L) \to a(K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} L)$ defined above for $K, L \in D_{QCoh}(\mathcal{O}_Y)$ is an isomorphism if K is perfect. In particular, (0A9S) is an isomorphism if K is perfect.*

Proof. Let $K^\vee$ be the "dual" to K, see Cohomology, Lemma 08DQ. For $M \in D_{QCoh}(\mathcal{O}_X)$ we have

$$\begin{aligned}\mathrm{Hom}_{D(\mathcal{O}_Y)}(Rf_*M, K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} L) &= \mathrm{Hom}_{D(\mathcal{O}_Y)}(Rf_*M \otimes^{\mathbf{L}}_{\mathcal{O}_Y} K^\vee, L) \\ &= \mathrm{Hom}_{D(\mathcal{O}_X)}(M \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*K^\vee, a(L)) \\ &= \mathrm{Hom}_{D(\mathcal{O}_X)}(M, Lf^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} a(L))\end{aligned}$$

Second equality by the definition of a and the projection formula (Cohomology, Lemma 0B54) or the more general Derived Categories of Schemes, Lemma 08EU. Hence the result by the Yoneda lemma. □